

ME 149 – Lab 2: Oscilloscope, Function Generator, and Multimeter

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Overview

In this experiment, I measured the oscillation of a simple pendulum using a smartphone's accelerometer and compared the results with theory. A smartphone was used to record acceleration during free oscillations. From the recorded data, I determined the pendulum's oscillation frequency and decay rate (Q). I finally plotted my results using MATLAB.

Equipment and Setup:

The experimental setup consisted of a smartphone attached to a string to act as a pendulum. Figures 1 and 2 show respectively a schematic and a picture of the setup. The following equipment was used:

Smartphone: Used its built-in accelerometer (via the Physics Toolbox app) to record acceleration in three axes.

String: Length $L = 0.5$ m, one end taped to the phone (inside a ziplock bag) and the other end clamped to a table to act as a pivot.

Clamp and Tape: To secure the string and phone in place and provide a stable pivot point.

Angle Measure: The phone was raised to a known initial angle, approximately 45° from vertical, before release.

MATLAB: The phone's app timestamped data, and MATLAB was used afterward to analyze the recorded data.

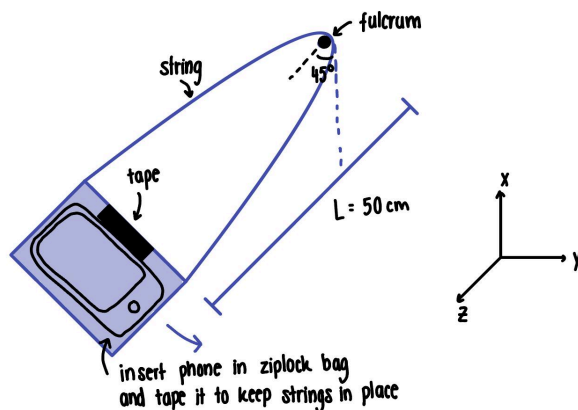


Figure 1. Pendulum's setup. The phone (with the accelerometer app running) was suspended as a pendulum and displaced about 45° from vertical, then released from rest to start oscillations. Acceleration data were recorded for about 4 minutes (~ 240 s) at a sampling rate of 100 Hz

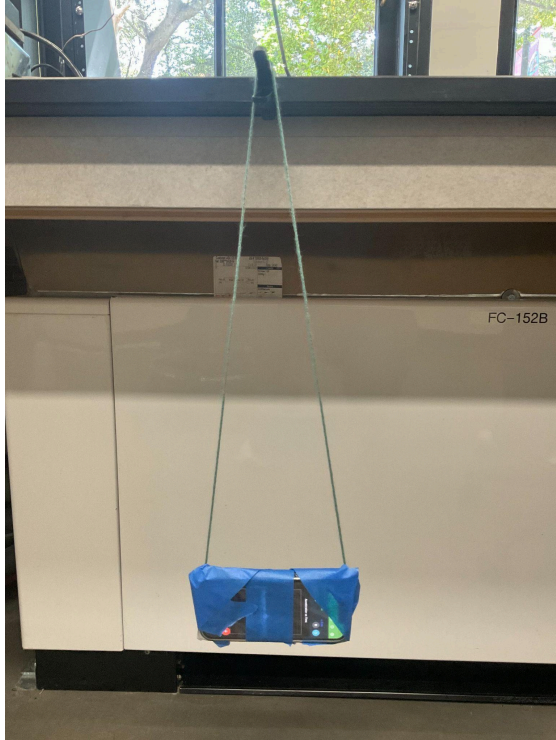


Figure 2: Image of experimental setup

Raw data

Figure 3 shows the x-axis acceleration vs time. The sampling rate was 1 reading every 0.01 seconds

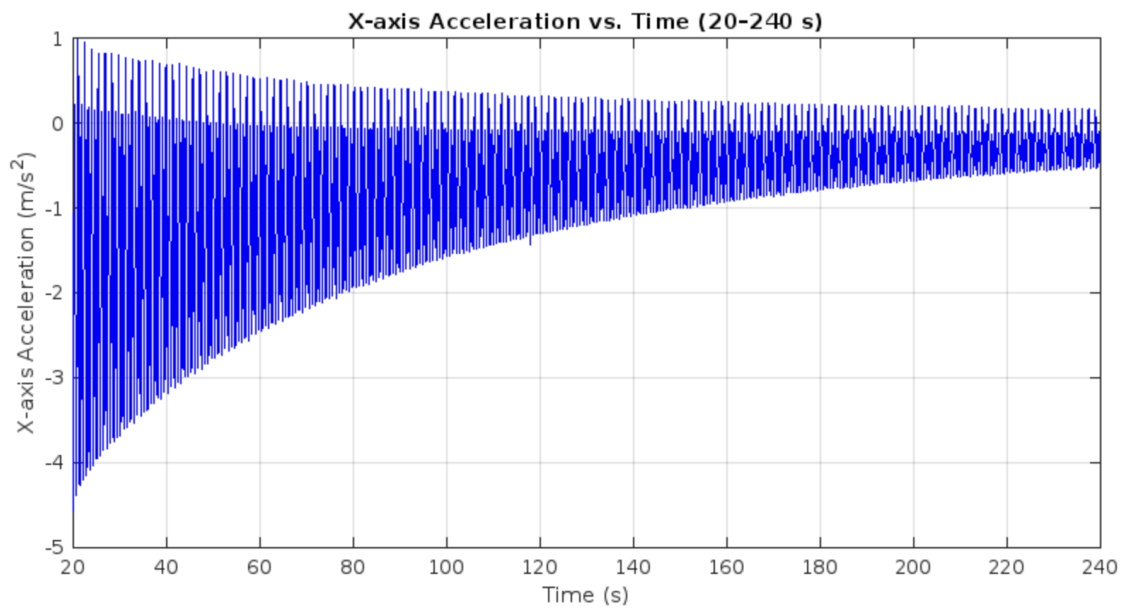


Figure 3: Plot of the x-axis acceleration vs time for ~4 minutes. This plot shows the acceleration in the width-wise direction of the phone in meters per second with a 100 Hz sampling rate

Figure 4 shows the y-axis acceleration vs time. The sampling rate was 1 reading every 0.01 seconds

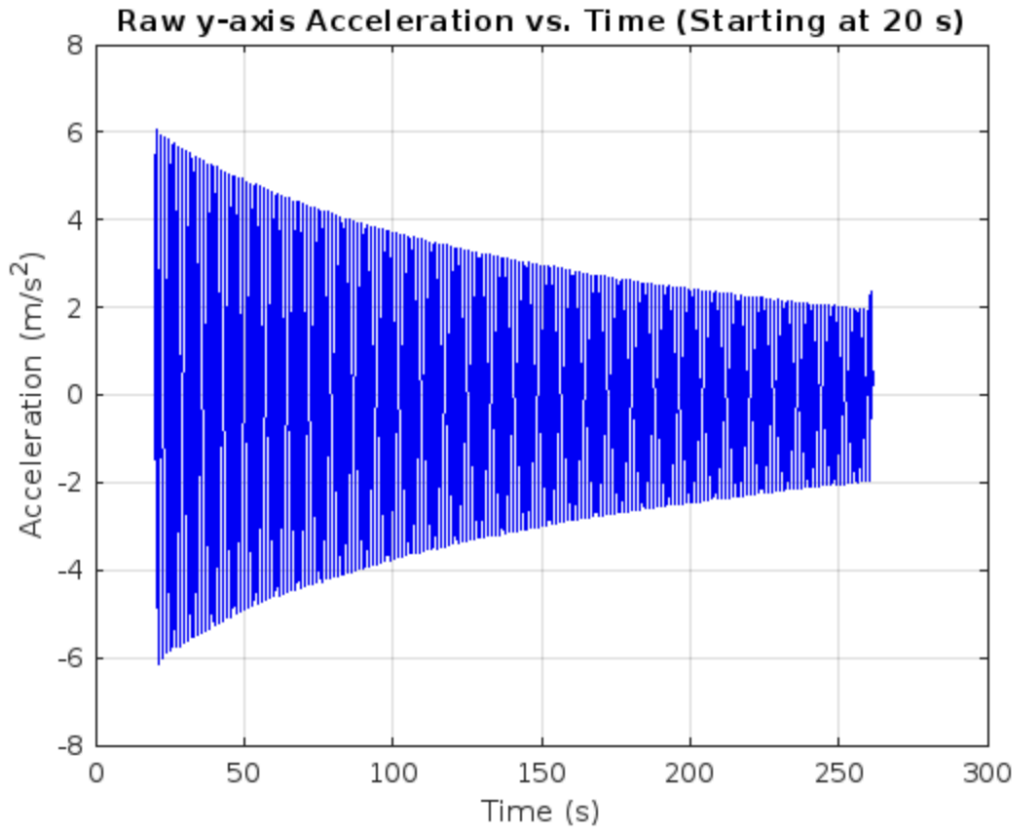


Figure 4: Plot of the y-axis acceleration vs time for ~4 minutes. This plot shows the acceleration in the width-wise direction of the phone in meters per second with a 100 Hz sampling rate

Figure 5 shows the z-axis acceleration vs time. The sampling rate was 1 reading every 0.01 seconds

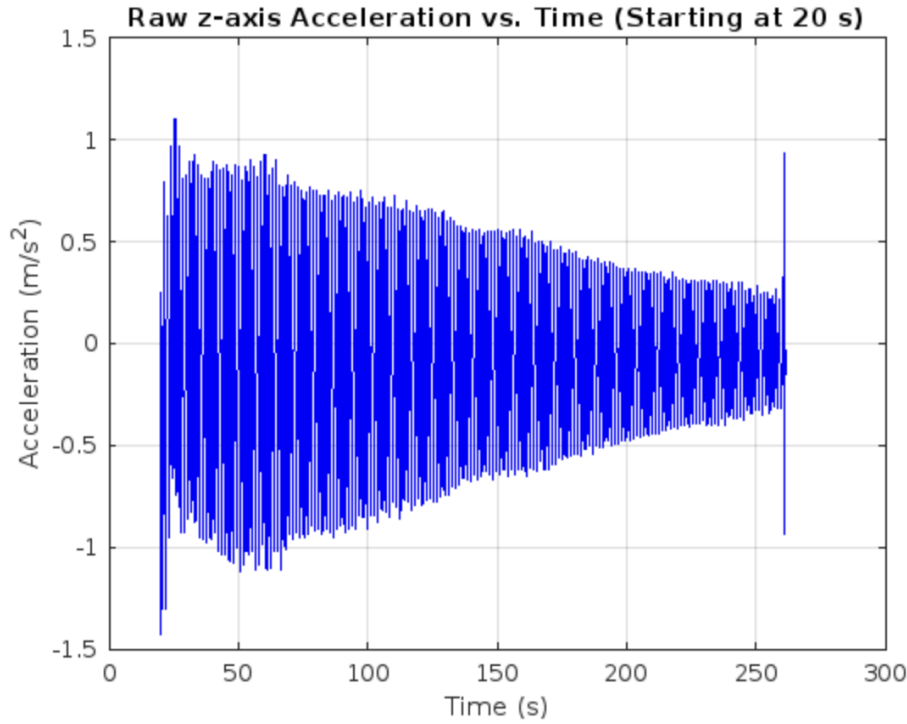


Figure 5: Plot of the z-axis acceleration vs time for ~4 minutes. This plot shows the acceleration in the width-wise direction of the phone in meters per second with a 100 Hz sampling rate

Data After Analysis:

Figure 6 plots the phone's lengthwise acceleration over the first 120 seconds of the run. By counting how many complete oscillation peaks occur in that 120 s window and dividing by 120 s, we obtain the experimental frequency. We then compare this measured value to the theoretical frequency given by:

$$f_{\text{theoretical}} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}, \quad f_{\text{estimated}} = \frac{\text{oscillations}}{\text{time}}$$

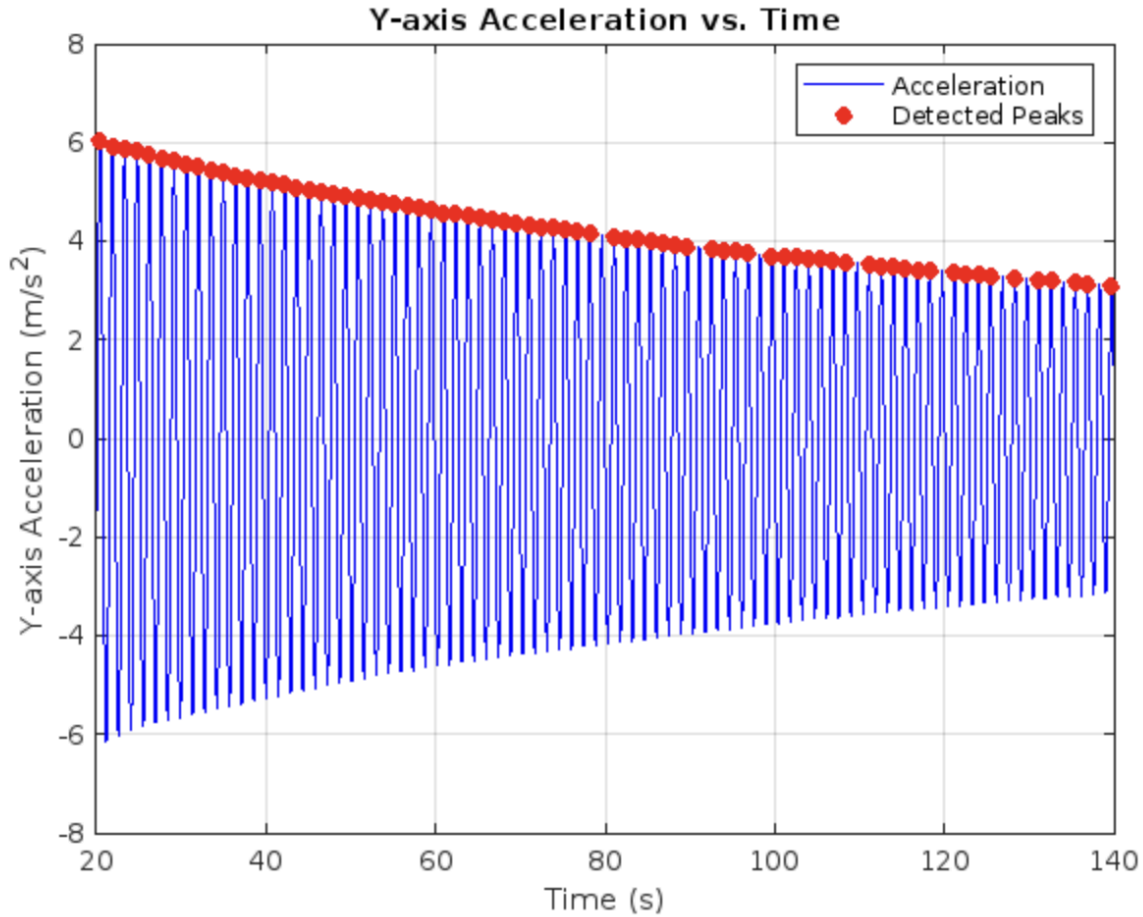


Figure 6: This graph shows the phone's lengthwise acceleration over the first two minutes of swinging, sampled at 100 Hz. By counting 75 oscillation peaks in that 120 s window, I determined the estimated oscillation frequency of 0.625 Hz. For a 0.5 m pendulum, the theoretical frequency predicts 0.704 Hz, which closely matches the my experiment.

Figure 7 plots the phone's lengthwise acceleration versus time and highlights the exact moment used to determine the pendulum's oscillation decay rate (the Q factor). The quality factor is calculated as $Q = N \cdot \pi$. N is the number of full swings required achieve *oscillation's initial maximum amplitude* / e.

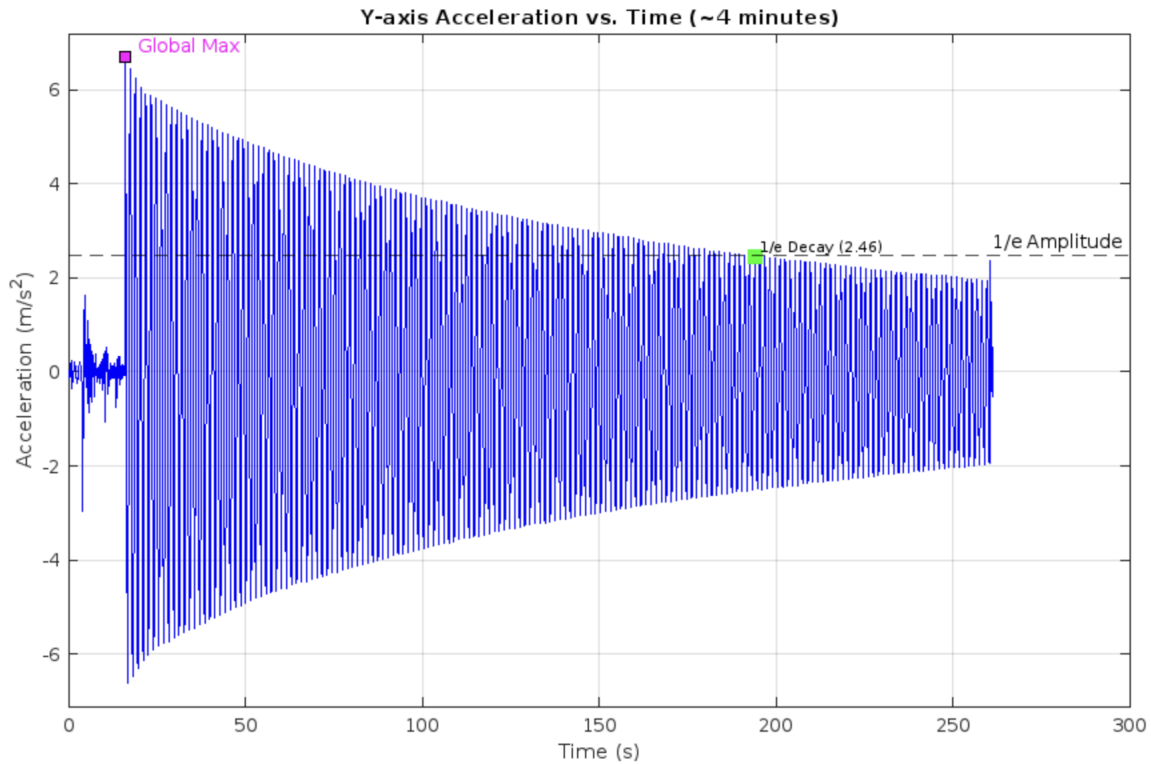


Figure 7: This plot shows the pendulum's lengthwise acceleration for the entire 4 minute measurement. The highest peak of 6.69 m/s^2 occurs at $t = 16.05 \text{ s}$ (Global Max). A dashed horizontal line shows the $1/e$ amplitude threshold of 2.46 m/s^2 . The green square marks when the oscillation first decays to that level at $t = 194.00 \text{ s}$, with an amplitude of 2.46 m/s^2 . Counting the number of full swings from the global maximum to this decay point gives $N = 135$ cycles, resulting in a quality factor of $Q = N\pi \approx 135 \pi \approx 424.12$.

In Figure 8, I show the power spectrum of the lengthwise acceleration data from Figure 6. To get this plot, I took the fast Fourier transform (FFT) of the signal in MATLAB and then divided the squared magnitude of the FFT by the total number of data points, N .

$$P = \frac{|FFT|^2}{N}$$

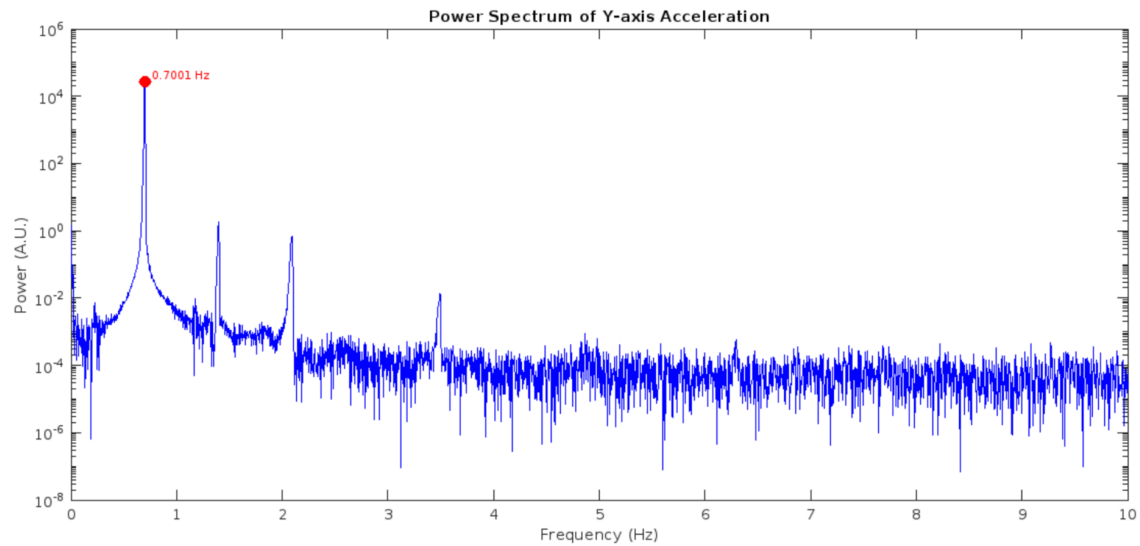


Figure 8: Plot of power spectrum of the pendulum's lengthwise (y-axis) acceleration, showing power on a log scale vs. frequency from 0 to 10 Hz. The largest peak in this plot is about **0.7001 Hz**. The FFT plot showed a peak at 0.700 Hz, which is very close to the theoretical 0.704 Hz for a 0.50 m pendulum.

Discussion

I used my phone's accelerometer and MATLAB to measure the pendulum's motion. From the first 120 s, counting 75 peaks gave me a frequency of 0.625 Hz. The FFT plot showed a peak at 0.7001 Hz, which is very close to the theoretical 0.704 Hz for a 0.50 m pendulum.

To find damping, I noted the largest acceleration (6.69 m/s^2) and saw it drop to 2.46 m/s^2 ($1/e$ of the max) at 194 s, after 135 swings. That gives a quality factor $Q \approx 424$, meaning the pendulum loses energy very slowly.